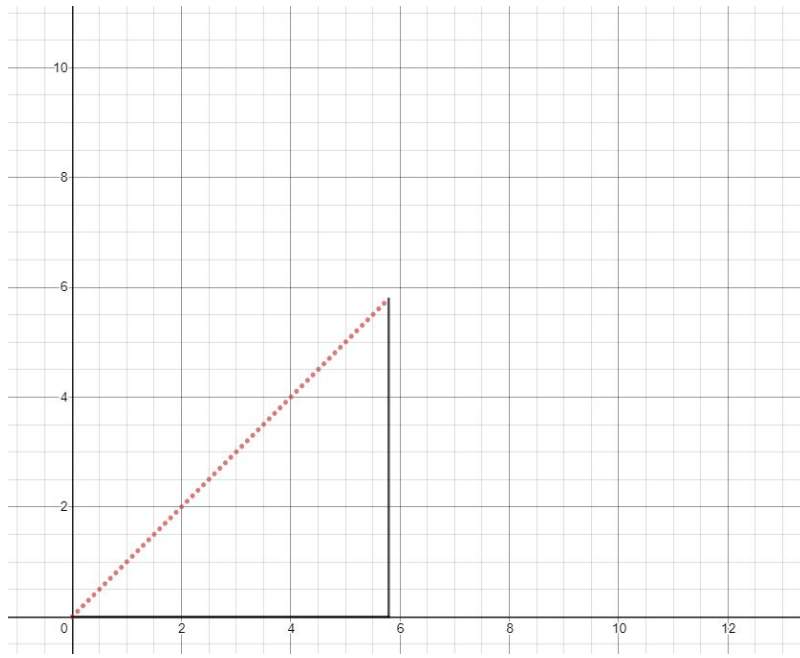
A unit vector is a vector with a magnitude of 1. To find the unit vector for a given vector, you simply divide the vector by its magnitude.

So, what would be the unit vector for $< 3, 4 >$? Well, since it's a Pythagorean triple, we know the magnitude is 5. Therefore, its unit vector would be
$< \frac{3}{5}, \frac{4}{5} >$.

$3i = < 3, 4 >$ or $3 < 1, 0 >$

$i = < 1, 0 >$. Unit vector parallel to the x-axis

$j = < 0, 1 >$. Unit vector parallel to the y-axis



Vectors also have **direction**. If the x-component is positive, it's going to the right. If it's negative, it's going to the left. If the y-component is positive, it's going upwards. If it's negative, it's going downwards. Pretty intuitive stuff.

The **component form** of a vector is the ordered pair that describes the changes in the x- and y-values.
For initial point (x_1, y_1) and terminal point (x_2, y_2), the component form is $< x_2 - x_1, y_2 - y_1 >$.

Vectors can be added, subtracted, multiplied, and divided with scalars (a number with no direction).

When adding or subtracting vectors, the x-components are added to each other and the y-components are added together.
If the 2 vectors were $\overline{v} = < 5, 6 >$ and $\overline{u} = < 6, 8 >$, then
$\overline{u} + \overline{v} = < 5, 6 > + < 6, 8 > = < 11, 14 >$. **THIS RULE WORKS FOR ADDING AND SUBTRACTING.**

Using the vectors from above for MULTIPLYING AND DIVIDING:

$2\overline{v} \quad = < 10,\ 12 >$

$\overline{u}/2 \ = < 3,\ 4 >$

IT IS ALSO IMPORTANT TO NOTE THAT VECTORS ARE NOT THE SAME AS RAYS.

When you multiply vectors, you use the **dot product**. You simply multiply the x's, multiply the y's, and then add them together. Your final answer should be <u>one single scalar value</u>:

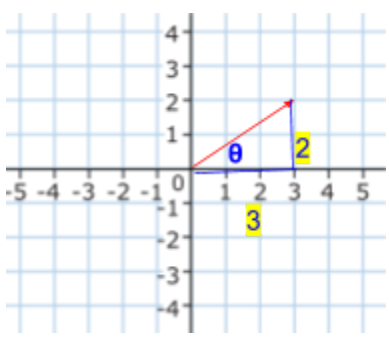
$< a_1,\ a_2 > \ \cdot \ < b_1, b_2 > \ = \ a_1 a_2 + b_1 b_2$

$< 2, 6 > \ \cdot \ < 12, 9 > \ = \ 24 + 54 \ = \ 78$

Direction angle is the angle the vector makes with the positive x axis.

To find the direction angle for a vector, we need to graph the vector on a coordinate plane and use trig to calculate the angle.

For example, to find the direction angle of the vector $\overline{v} \ = \ < 3,\ 2 >$, we make the following graph:



To find the angle the vector makes with the x axis, we need to use tangent:

$\theta = tan^{-1}(2/3) = 33.69°$

Conics

They occur frequently in nature. It's still an equation that relates two variables. It's just that y is not a function of x because the two variables are related in such a way that you can't make one the dependent variable and the other the independent variable.

<u>Circles</u>

Let's start from the basics. You know what a circle is. Every point on it is equally distant from the center. Where the point (h, k) is the center of the circle and r is the radius, this is the standard form equation for a circle.

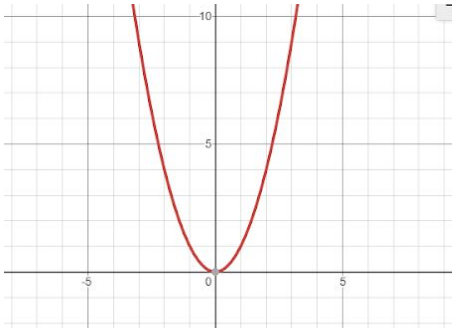
$$(x-h)^2+(y-k)^2=r^2$$

If the equation is given to you in standard form, it's pretty easy to graph. Identify the center and the radius and it should be self-explanatory from there. For example, if you're graphing $(x-4)^2+(y+3)^2=16$, then you'd plot the center point at (4, -3) and go out 4 units from all sides of the center.

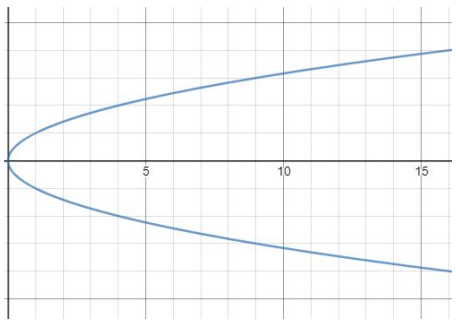
The other skill you'll need to be successful in this unit is converting other forms to standard form. To do this, the most important thing is completing the square. Refer to this concept for a more in depth explanation.

Parabolas

The conics unit expands on what you already know about quadratics. As you know, parabolas can look like this:



But, they can also be sideways, like this:



There's a lot more to delve into with parabolas. First, we should look at the new and improved formulas that you will need to know.

The equations are:

$$x = \frac{1}{4p}(y-k)^2 + h \quad \text{(sideways)}$$

$$y = \frac{1}{4p}(x-h)^2 + k \quad \text{(upright)}$$

What do all these variables mean? Just like with other transformation formulas, h and k represent horizontal and vertical shifts, respectively. Additionally, the point (h,k) is known as the vertex.

For example, if we had the following parabola:

$$y = (x-1)^2 + 1$$

then the vertex would be located at (1,1).

Another part of this formula is the front coefficient $\frac{1}{4p}$ which involves the variable p. The **direction** of the parabolas depends on the isolated variable (x or y) and the leading coefficient (positive or negative).

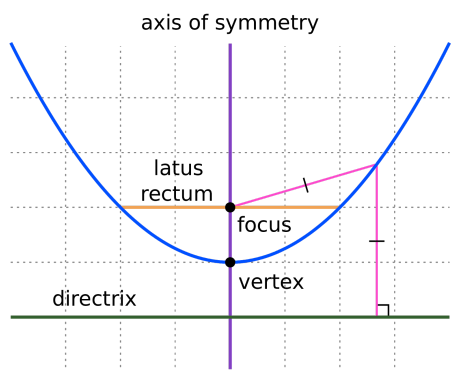
For example, if the isolated variable is **x**, then the graph will open **left or right**. If the isolated variable is **y**, then the graph will open **up and down**. If the leading coefficient is **positive**, then the graph will either open **up or right**. If the leading coefficient is **negative**, the graph will open **down or left**.

The **focal length**, or the distance between the vertex and focus, is the value of **p** in $\frac{1}{4p}$.

The **focus** is the point you get when you move **p** units from the vertex of the parabola towards the inside of the parabola. (see graph)

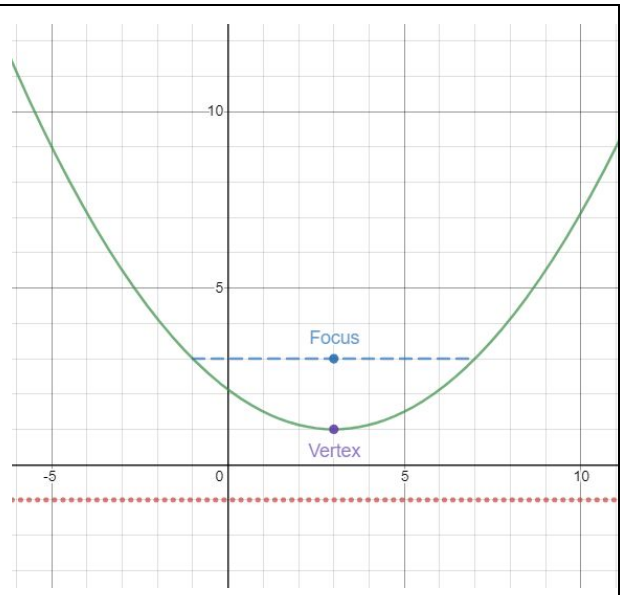
The **directrix** is the line you get when you move **p** units from the vertex of the parabola towards the outside of the parabola. (see green line on the graph)

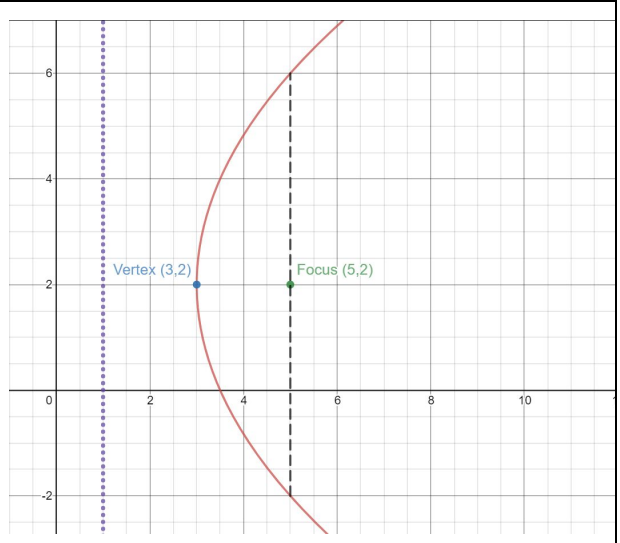
The **focal width** is equal to the value of **4p.** It is a line that extends from one side of the parabola to the other side, passing through the focus. (see orange line on the graph)



Here's some examples:

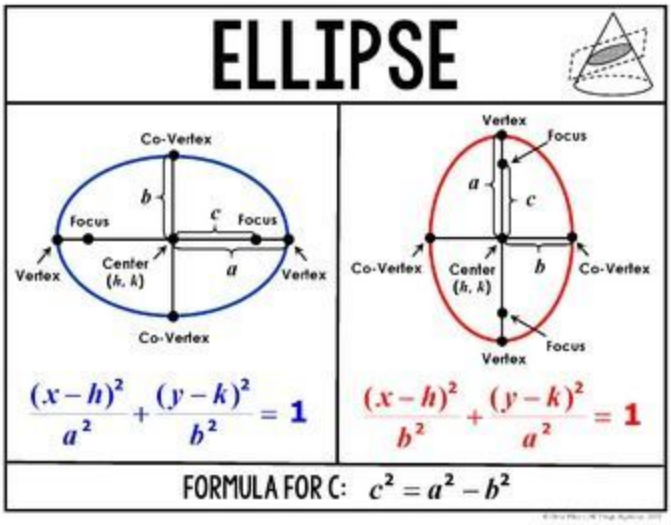
Example 1:

$y = \frac{1}{8}\ (x-3)^2 + 1$			
Opens **up**	Vertex **(3,1)**		
Focus **(3,3)**	Directrix: **y=-1**		
Focal length **p=2**	Focal Width **4p=8**		
Domain **(-∞ , ∞)**	Range **[1, ∞)**	X-int: none	Y-int: $(0, \frac{17}{8})$

Example 2:			
$x = \frac{1}{8}\ (y-2)^2\ + 3$			
Opens: **right**	Vertex: **(3,2)**		
Focus: **(5,2)**	Directrix: **x= 1**		
Focal length: p=**2**	Focal Width: 4p= **8**		
Domain: **[3, ∞)**	Range: **(-∞ , ∞)**	X-int: **(3.5, 0)**	Y-int: **none**

Ellipses

Ellipses are like stretched circles/ovals. The planets in our solar system have elliptical orbits around the sun. Ellipses can be horizontal or vertical, depending on whether its x length or y length is longer.



It has two vertices and two co-vertices. These are the points at the end of the axes. Refer to the picture on the right for clarification. There are also two foci along the major axis length, which is the longer length. The distance between the foci and any point on the ellipse is always the same.

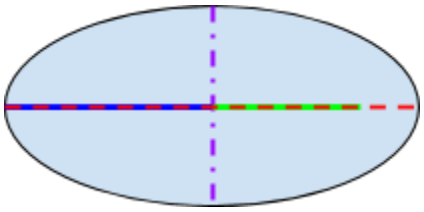
The standard form for a horizontal ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

The standard form for a vertical ellipse is $\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$.

Yes, the only difference is the a and the b are swapped. Why? We're using a to represent the longer length.

We determine if an ellipse is horizontal or vertical by seeing which variable is above the bigger denominator, or a^2. The center of the ellipse is (h, k). You should also determine the values of a and b by square rooting the denominators.

With a, you can find the major axis length and the major vertices. The major axis is the longer length of the ellipse. In this ellipse on the right, the red dotted line represents the major axis. The solid blue line represents a. Thus, to find the major axis length, you just multiply a by 2. To find the vertices, you just add and subtract a from the center.



This same process is used for the minor axis and vertices.

The last major parts of the ellipses are the foci. The length from the center to one focus is represented with c. In the pictured ellipse, c is the neon green line. To find the c, use the formula $c^2 = a^2 - b^2$.

Looks take a look at an example.

$$\frac{(x-3)^2}{16} + \frac{(y+6)^2}{4} = 1$$

1. Horizontal or vertical: Since x is over the larger denominator, the ellipse is **horizontal**. Remember to square root the denominators to find a and b. We know that $a = 4$ and $b = 2$.
2. Center: (3, -6)
3. Major axis vertices: We know the ellipse is horizontal, so the major side is the x length. That means we're dealing only with x and a when finding the vertices. The two vertices will go 4 left and 4 right from the center, so they are (-1, -6) and (7, -6).
4. Minor axis vertices: Now it's the y. You go up 2 and down 2, so the vertices are (3, -8) and (3, -4).
5. Major axis length: $2a = 8$
6. Minor axis length: $2b = 4$
7. Foci: $c^2 = a^2 - b^2$
   a. $c^2 = 4^2 - 2^2$
   b. $c^2 = 16 - 4 = 12$
   c. $c = \pm 2\sqrt{3}$
   d. Since foci are along the major axis, we will add and subtract c from the x component of the center. Therefore, the two foci are $(3 + 2\sqrt{3}, 6)$ and $(3 - 2\sqrt{3}, 6)$.
8. Domain: [-1, 7] (You can refer to your vertices for domain and range!)
9. Range: [-8, -4]

As always, if you want to find x-intercepts, set y = 0. If you want to find y-intercepts, set x = 0.

Okay, how do you convert a non-standard form ellipse into standard form? You'll need to complete the square, but you have to be careful when you do so. Sometimes, the math can get tricky if you don't pay close attention.

Here are the steps:

1. Group variables
2. Factor out LCM
3. Complete the square

4. Rewrite
5. Put into standard form

Here's the example:

$$9x^2 + 4y^2 - 18x + 8y - 23 = 0$$

$9x^2 - 18x + 4y^2 + 8y = 23$ *Group variables

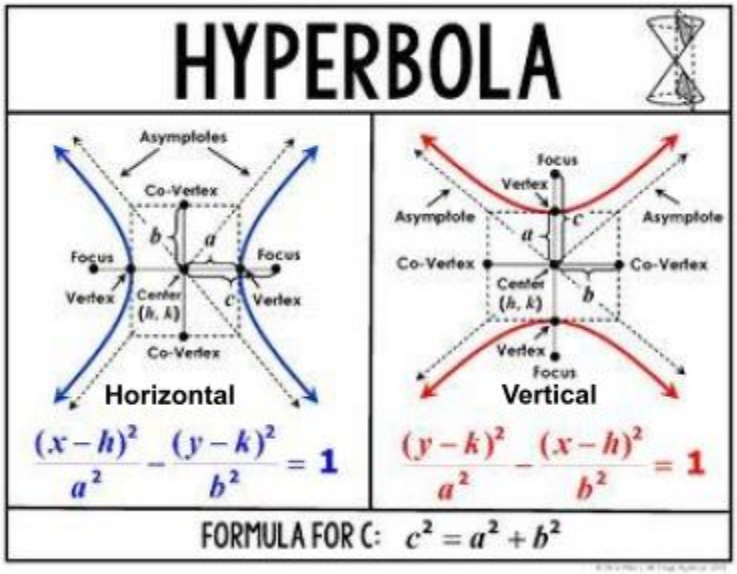
$9(x^2 - 2x) + 4(y^2 + 2y) = 23$ *Factor out LCM

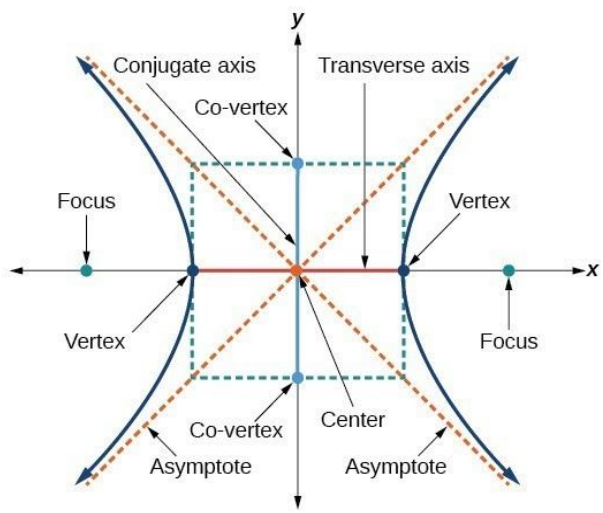
$9(x^2 - 2x + 1) + 4(y^2 + 2y + 1) = 36$ *Complete the square (Note that when you're adding a 1 into the parenthesis, you're actually adding a 9 overall due to the coefficient out front. Be careful about this! That is why 9 and 4 were added to 23 even though only two "1"s were put into the left side.

$9(x - 1)^2 + 4(y + 1)^2 = 36$ *Rewrite

$\frac{(x-1)^2}{4} + \frac{(y+1)^2}{9} = 1$ *Put into standard form (The equation needs to equal 1, so every term was divided by 36)

Hyperbola: these look kind of like two parabolas facing away from each other





The formula for hyperbolas are:

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1 \text{ and } \frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

This is the standard form equation.

If the **x** comes first in the equation, the graph is **horizontal,** looking like two C's facing outwards.
If the **y** comes first, the graph is **vertical,** looking like two U's facing outwards.

The **center** of a hyperbola is (h,k).

Each **asymptote** will be a linear equation. The equation for the asymptote is $y = \pm \frac{\Delta y}{\Delta x}(x-h) + k$

$\frac{\Delta y}{\Delta x}$ is the value under the y divided by the value under the x

The **foci** are two points located inside the curves of the hyperbola (see "focus" on the pictures)
The formula for **foci** or the **c** value is $c^2 = a^2 + b^2$
To find the **foci points**, you add or subtract **c** from the **x value** (horizontal hyperbola) **or y value** (vertical hyperbola) of the center point

The **vertices** are the 2 points in the center of each curve (see "vertex" on picture)
To find the vertices, we move left/right (for horizontal hyperbolas) and up/down (for vertical hyperbolas) **"a"** units from the center.

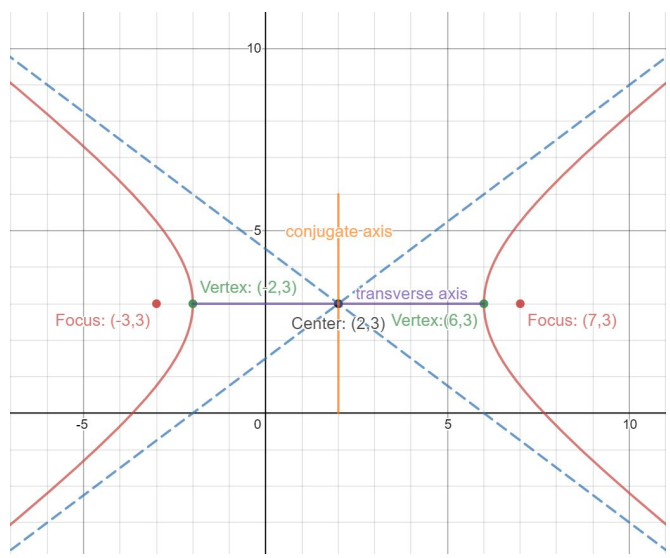
The **vertices** are also the endpoints of the **transverse axis.**

If you move left/right or up/down from the both the **vertices "b"** units, you will get 4 points. If you connect the 4 points diagonally, you will have drawn the **asymptotes** of the hyperbola.

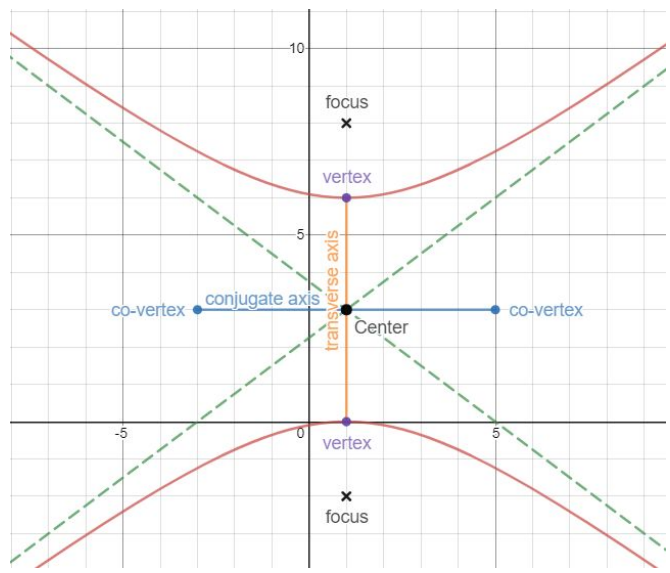
The **transverse axis length** is equal to **"2a"**
The **conjugate axis length** is equal to **"2b"**

<u>To graph the hyperbola, you draw through the vertices approaching the asymptotes.</u>

Here is an example problem:

<table>
<tr><td colspan="2">$\frac{(x-2)^2}{16} - \frac{(y-3)^2}{9} = 1$
(you can see that the a value is 4 and the b value is 3)</td><td rowspan="4"> 
</td></tr>
<tr><td>Horizontal or Vertical?: horizontal</td><td>Center: (2,3)</td></tr>
<tr><td>Vertices: (-2,3) and (6,3)
(we move 4 units left and right of the center)</td><td>Asymptotes:
$y = \pm \frac{3}{4}(x-2) + 3$</td></tr>
<tr><td>Transverse Axis length: 8</td><td>Conjugate axis length: 6</td></tr>
<tr><td colspan="2">Foci:
$c^2 = a^2 + b^2$</td><td></td></tr>
</table>

$c^2 = 4^2 + 3^2$ $c^2 = 25$ so c=5 **(2 ± 5, 3) or (-3,3) and (7,3)**			
Domain: **(-∞ , -2] U [6, ∞)**	Range:**(-∞ , ∞)**	X-int: $(\pm\sqrt{32} + 2, 0)$ $\frac{(x-2)^2}{16} - \frac{(-3)^2}{9} = 1$ $\frac{(x-2)^2}{16} - 1 = 1$ $\frac{(x-2)^2}{16} = 2$ $(x-2)^2 = 32$ $x - 2 = \pm\sqrt{32}$ $x = \pm\sqrt{32} + 2$	Y-int: **NA**

$\frac{(y-3)^2}{9} - \frac{(x-1)^2}{16} = 1$ (you can see that the a value is 3 and the b value is 4)			
Horizontal or Vertical?: **vertical**	Center: **(1,3)**		
Vertices: **(1,6)** **(1,0)**	Asymptotes: $y = \pm\frac{3}{4}(x-1) + 3$		
Transverse Axis length: **6**	Conjugate axis length: **8**		
Foci: **(1,8) (1,-2)**			
Domain: **(−∞, ∞)**	Range: **(−∞, 0]∪[6,+∞)**	X-int: **(1,0)** *when* $y = 0$, $\frac{(0-3)^2}{9} - \frac{(x-1)^2}{16} = 1$ $\frac{9}{9} - \frac{(x-1)^2}{16} = 1$ $-\frac{(x-1)^2}{16} = 0$ $(x-1)^2 = 0$ $x = 1$	Y-int: $(0, 3 \pm \sqrt{\frac{153}{16}})$ *when* $x = 0$, $\frac{(y-3)^2}{9} - \frac{(0-1)^2}{16} = 1$ $\frac{(y-3)^2}{9} - \frac{1}{16} = 1$ $\frac{(y-3)^2}{9} = \frac{17}{16}$ $(y-3)^2 = \frac{153}{16}$

			$y - 3 = \pm\sqrt{\frac{153}{16}}$ $y = 3 \pm \sqrt{\frac{153}{16}}$

Cheat Sheet:

Ellipses

Formulas:

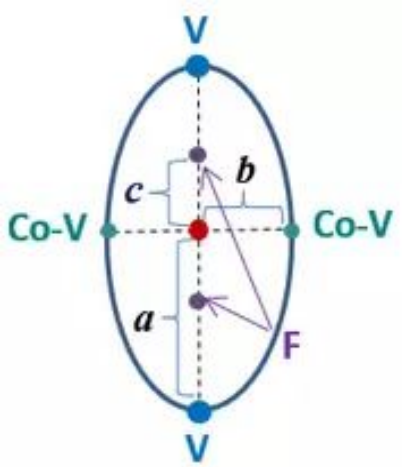
Horizontal: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

Vertical: $\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$

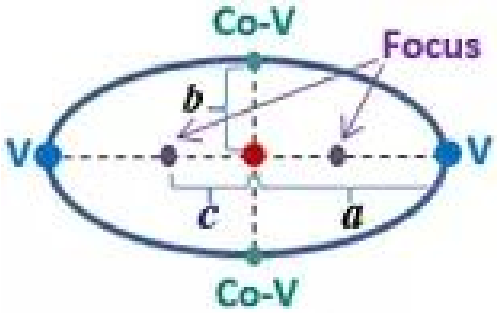
Major axis length: $2a$
Minor axis length: $2b$

$c^2 = a^2 - b^2$

Vertical



Horizontal

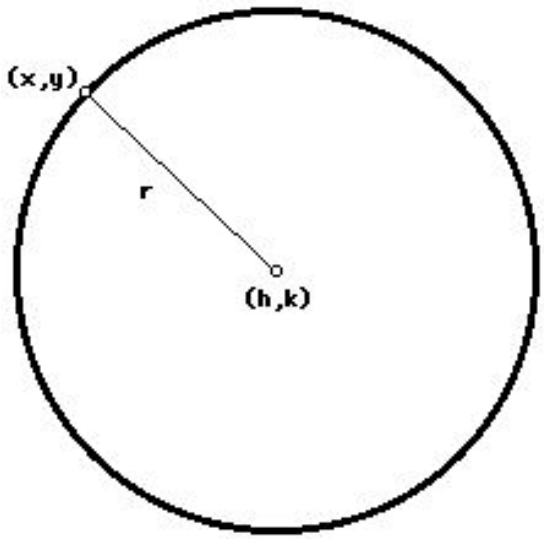


Circles

Formulas

$$(x-h)^2 + (y-k)^2 = r^2$$

h is horizontal shift and x coordinate of vertex
k is vertical shift and y coordinate of vertex
r is radius



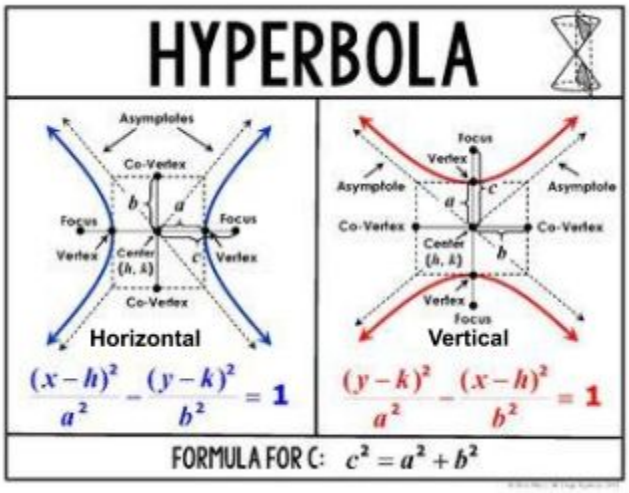
Hyperbolas

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1 \text{ and } \frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

Formulas

$c^2 = a^2 + b^2$

TRANSVERSE axis length: 2a
CONJUGATE axis length: 2b



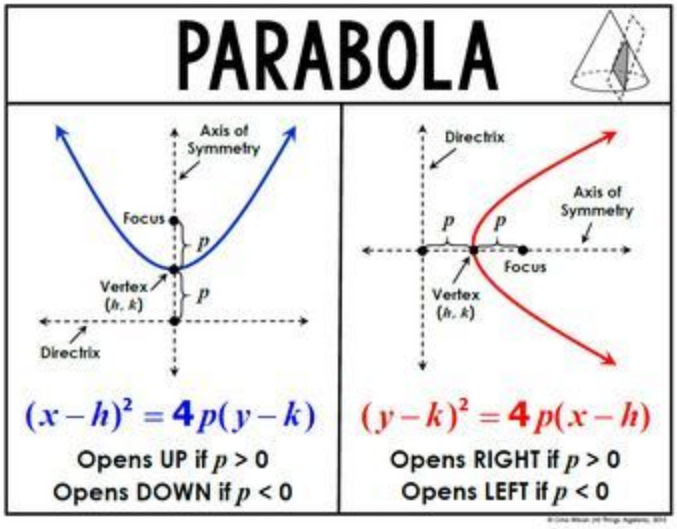
Parabolas

Formulas

$x = \frac{1}{4p}(y-k)^2 + h$ (sideways)

$y = \frac{1}{4p}(x-h)^2 + k$ (upright)

focal length=p

focal width=4p

directrix= line p away from the vertex, opposite direction from the focus